Rodrigo Rocha

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SUMMARY

Final-year Informatics and Computer Engineering student at Faculdade de Engenharia da Universidade do Porto, already taking master's units, focused on backend and full-stack development. Built an end-to-end document processing pipeline in Java/Spring Boot, PostgreSQL and Angular at SolvIt, and modelled FEUP's exam scheduling in IBM ILOG CPLEX, cutting same-day clashes by 88.9% on real data. Looking for backend or full-stack roles.

EXPERIENCE

SolvIt, Jul. 2026 – Aug. 2026
Software Engineering Intern Porto, Portugal

- Built an end-to-end document processing pipeline in Java and Spring Boot, enabling automated ingestion and classification of PDFs and images.
- Extracted structured fields from documents and implemented SHA-256 deduplication with content-based validation to ensure data integrity.
- Designed the persistence layer in PostgreSQL, using Flyway migrations for versioned schema changes and normalising pages for consistent document storage.
- Developed the Angular front-end for document upload and review of extraction results, consuming the back-end through REST APIs.

PROJECTS

Exam Timetabling Optimization Engine — Capstone Project, FEUP Feb. 2026 – Jul. 2026
IBM ILOG CPLEX, Mixed-Integer Programming, Python Porto, Portugal

- Formalised FEUP's exam scheduling rules as a combinatorial optimization model in IBM ILOG CPLEX.
- Built a Python pipeline scraping real enrolment data from the university's academic system for the full Informatics Engineering programme.
- Reduced same-day exam clashes by 88.9% and consecutive-day exams by 78.1% against the official timetable.

EDUCATION

Faculdade de Engenharia da Universidade do Porto, Sep. 2023 – 2026
Bachelor's in Informatics and Computing Engineering Porto, Portugal

- Currently completing the BSc while already taking master's (M.EIC) curricular units in parallel.
- Coursework: Algorithms and Data Structures, Databases, Software Engineering, Computer Networks, Compilers, Artificial Intelligence, Machine Learning.

Escola Secundária Vitorino Nemésio Concluded 2023
Secondary Education — Science and Technology Praia da Vitória, Portugal

ACTIVITIES

Tuna de Engenharia da Universidade do Porto, Sep. 2023 – Current
Performer & External Relations Porto, Portugal

- Manage external booking correspondence for a 60-member ensemble, including pricing negotiations, contract terms and logistics with festivals, municipalities and private clients.
- Perform at concerts, festivals and private events across Portugal and abroad, contributing to cultural exchange and community engagement.

Centro de Eventos FEUP, Jul. 2025 – Current
Audiovisual Technician Porto, Portugal

- Set up, operate and dismantle audio, video, streaming and lighting systems for conferences and congresses, troubleshooting issues in real time during live sessions.

Mentoria U.Porto, Sep. 2025 – Current
Mentor Porto, Portugal

- Support new engineering students through peer mentoring, adapting each programme to the mentee's needs.

SKILLS

Programming Languages: Java, Python, C++, C, TypeScript/JavaScript, SQL, PHP

Backend & Frontend: Spring Boot, JPA/Hibernate, Node.js, REST APIs, Angular, React, HTML/CSS

Databases & Infrastructure: PostgreSQL, MySQL, Flyway, Docker, Git, Maven/Gradle, Linux

Optimization & Systems: IBM ILOG CPLEX, Mixed-Integer Programming, compiler design (LLVM-style IR), WebGL/OpenGL

Languages: Portuguese (native), English (fluent)